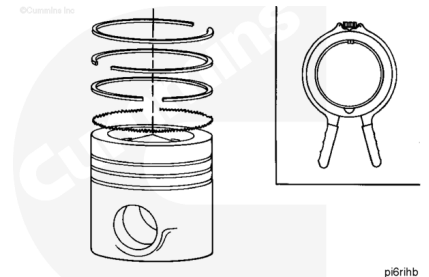


Trainings ▾

Remove

Use piston ring expander, Part Number ST-1269, or equivalent.

Remove the piston rings.

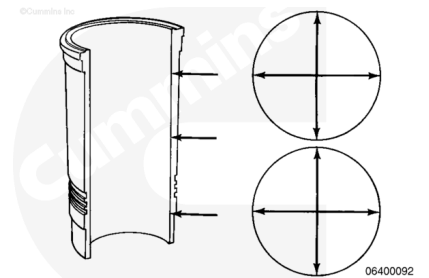


Measure

with Mechanically Actuated Injector

Make certain the liners are within specification prior to measuring piston ring gaps.

- For the K19 engine, use the following procedure in K19 Service Manual, Bulletin 4021499. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html)
- For the QSK19 engine, use the following procedure in QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)



Note : Once a set of piston rings has been measured against a cylinder liner, they must only be used on a piston which is to be inserted into that liner. If the piston rings are subsequently required to be used in another cylinder location, the ring gap measuring process described here must be carried out again in the new cylinder liner.

To check the ring gap, place an ink mark on the side of the serviceable piston (with no piston rings fitted) which is 100 mm [3.94 in] from the crown.

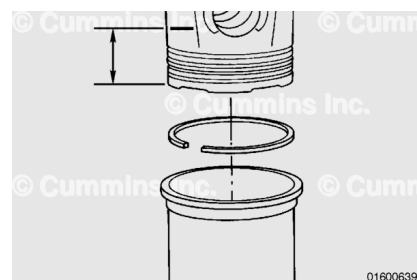
Install a piston ring in the top of a cylinder liner that meets specifications and will be used with the piston rings having their gaps measured.

Use the top part of the piston to position the ring in the liner. Drop the piston down until the 100 mm [3.94 in] marking on the piston reaches the top of the liner.

Carefully remove the piston and carry out the ring gap measurement.

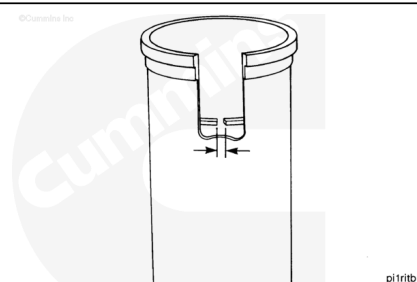
Note : Engines built after December 2008 have a revised top piston ring. For these engines, the top piston ring gap **must** be measured using the specification for an electronically actuated injector.

Note : Any engine built prior to December 2008 may have the same top piston ring used in electronically actuated injector pistons, or they may have the top piston ring used in the mechanically actuated injector piston. Check the part number on the top piston ring to make sure the correct tolerances are used for the piston ring gap.



⚠ CAUTION ⚠

If the ring gap is not within the minimum specification, do not attempt to adjust the end gap by filing or grinding. The chrome plating on the ring will be damaged and the ring will malfunction. Check to make sure the cylinder bore inside diameter is correct. If the end gap remains less than the minimum specification, replace the piston ring.



Use a feeler gauge to check the piston ring gap.

The ring **must** be replaced if it does **not** meet the following specifications.

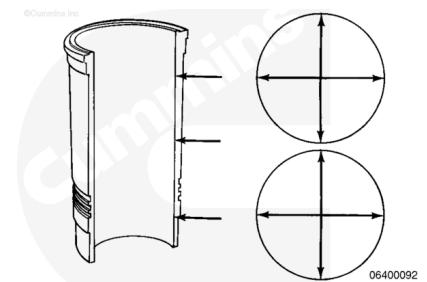
Piston Ring Gap - Mechanically Actuated Injector			
	mm		in
Top	0.45	MIN	0.018
	0.65	MAX	0.026
Intermediate	0.90	MIN	0.035
	1.20	MAX	0.047

Oil	0.38	MIN	0.015
	0.76	MAX	0.030

with Electronically Actuated Injector

Make certain the liners are within specification prior to measuring piston ring gaps.

- For the K19 engine, use the following procedure in K19 Service Manual, Bulletin 4021499. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/18/18-001-028-tr.html)
- For the QSK19 engine, use the following procedure in QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)



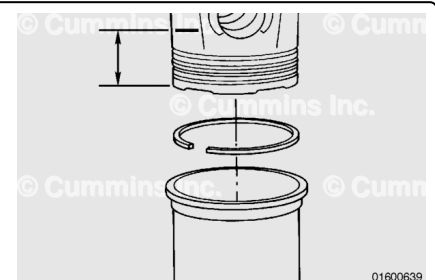
Note : Once a set of piston rings has been measured against a cylinder liner, they **must only** be used on a piston which is to be inserted into that liner. If the piston rings are subsequently required to be used in another cylinder location, the ring gap measuring process described here **must** be carried out again in the new cylinder liner.

To check the ring gap, place an ink mark on the side of the serviceable piston (with no piston rings fitted) which is 100 mm [3.94 in] from the crown.

Install a piston ring in the top of a cylinder liner that meets specifications and will be used with the piston rings having their gaps measured.

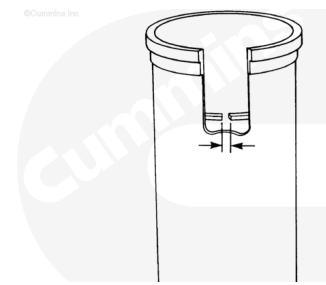
Use the top part of the piston to position the ring in the liner. Drop the piston down until the 100 mm [3.94 in] marking on the piston reaches the top of the liner.

Carefully remove the piston and carry out the ring gap measurement.



⚠ CAUTION ⚠

If the ring gap is not within the minimum specification, do not attempt to adjust the end gap by filing or grinding. The chrome plating on the ring will be damaged and the ring will malfunction. Check to make sure the cylinder bore inside diameter is correct. If the end gap remains less than the minimum specification, replace the piston ring.



Use a feeler gauge to check the piston ring gap.

The ring **must** be replaced if it does **not** meet the following specifications.

Piston Ring Gap - Electronically Actuated Injector			
	mm		in
Top	0.63	MIN	0.025
	1.02	MAX	0.040
Intermediate	0.90	MIN	0.035
	1.20	MAX	0.047
Oil	0.38	MIN	0.015
	0.76	MAX	0.030

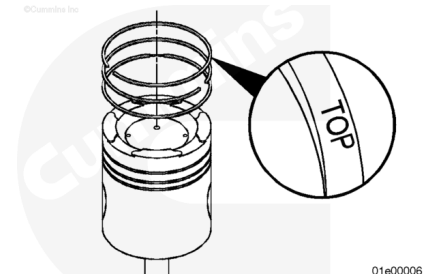
Install

Note : Do not install the piston rings until the piston is installed onto the rod. For the K19 engine, use the following procedure in K19 Service Manual, Bulletin 4021499, to assemble the piston to connecting rod. Refer to Procedure 001-054 in Section 1.

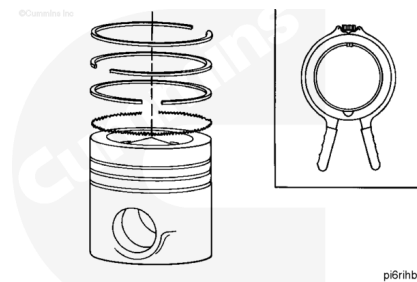
(/qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html) For the QSK19 engine, use the following procedure in QSK19, QSK19 CM850 MCRS, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592, to assemble the piston to connecting rod. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html)

The top side of the piston rings are marked with the part number. The shipping package is marked with the location. The part number on the ring **must** be positioned toward the top of the piston.



Use a ring expander, Part Number ST-1269, or equivalent, to install the piston rings.



Do **not** align the gap of a ring with the piston pin bore. Rotate the rings until the gaps are positioned as shown.

The gap in the oil ring expander (2) **must** be turned 180 degrees opposite the gap on the oil ring (1).

Position the remaining ring gaps approximately 120 degrees apart from the oil ring gap, as shown in the illustration.

